

Leading University

Department of Computer Science and Engineering

Course Code: 3240



MediMind

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28th February 2026

MediMind



A Project submitted to the
Department of Computer Science and Engineering
Leading University

Sylhet - 3112, Bangladesh

in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science and Engineering

By

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Recommendation Letter from the Project Supervisor

The project entitled "*MediMind*" submitted by the students

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This is a record of research work carried out under my supervision, and I hereby approve that the report be submitted in partial fulfillment of the requirements for the award of their bachelor's degrees.

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Certificate of Acceptance of the Project

The project entitled *MediMind*
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is hereby accepted as the partial fulfillment of the requirements for the award of their
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Abstract

In today's fast-paced environment, consistent medication adherence is often overlooked, even though it is essential for effective treatment and long-term health. With missed doses and complex schedules leading to rising health complications, MediMind is a mobile-based initiative designed to offer a reliable, intuitive, and empowering digital space for personal healthcare management. The app integrates three key modules: seamless medication management for adding and deleting prescriptions, timely push notifications for scheduled doses, and a dedicated oversight feature that allows a primary user to monitor the adherence of up to five dependents or loved ones. MediMind balances structured medical routines with collaborative family support. Powered by reliable scheduling algorithms and real-time database synchronization, the app ensures accurate dosing alerts, tracks user compliance, and fosters accountability through shared monitoring. By bridging the gap between isolated treatments and collective care, MediMind nurtures a culture of health consciousness, safety, and mutual support within families and caregiving circles.

Index Terms— Medication Adherence; Mobile Health; Remote Monitoring;

Acknowledgment

We would like to express our sincere gratitude to the Department of Computer Science and Engineering, Leading University, Sylhet, for providing the resources and support necessary to complete this coursework. Special thanks are also extended to our supervisor, **Md. Jamaner Rahman** sir, for his immense support, invaluable guidance, and constant encouragement throughout the duration of this project.

Dedication

We dedicate this project to our parents and siblings, whose constant encouragement, support, and belief in our abilities have been instrumental to our growth. We also extend this dedication to all individuals who diligently manage their health challenges, the caregivers who provide silent support, and those who strive for a better quality of life. Medimind is built for them, with the hope of fostering reliability, peace of mind, and better health outcomes within our community.

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Chapter 1: Introduction

Medimind is a mobile application designed to enhance medication adherence and health management by providing a reliable, inclusive, and user-friendly digital platform for patients and caregivers. Developed using Flutter and Dart, with Firebase serving as the backend for authentication and database management, Medimind enables users to schedule precise medication reminders, track their dosage history, and monitor health records seamlessly.

The app integrates a specialized notification system and a centralized dashboard, allowing users to receive real-time alerts and track compliance patterns. Unlike generic reminder tools, Medimind is localized and tailored specifically for the needs of our community, offering a dual-mode interface for both patients and those overseeing their care. Through a seamless blend of technology and reliability, the system encourages health accountability, reduces the risk of missed doses, and supports better medical outcomes in a secure and accessible space.

Chapter 2: Problem Description

2.1 Problem Statement

In recent years, medication non adherence has become a growing concern within healthcare systems, particularly for elderly and ill individuals who must manage multiple prescriptions daily. Despite the availability of generic health applications, most are either too complex for older users or lack the specific features needed to connect patients with their caregivers effectively. They often fail to address the practical challenges of managing long term treatments, where a lack of organization can lead to serious health risks. The primary goal of this project is to design and develop a mobile application named Medimind, which provides a reliable and secure digital platform for medication tracking and health accountability. The system focuses on creating a user friendly space for individuals to schedule precise reminders and manage their medication schedules seamlessly. Our team task was to conceptualize, design, and implement this application using Flutter for cross platform development and Firebase as the backend for authentication and data management. Additionally, the system features a dual mode interface that allows for both a Patient Mode and a Monitor Mode, ensuring that caregivers or family members see real time updates on the status of a user. The prototype is developed primarily for our university community and their families, specifically targeting elderly or ill people who require consistent support. It is designed to operate within limited hardware resources such as Android smartphones and utilizes the Firebase backend to ensure smooth performance and data privacy.

2.2 Goals

The Medimind project aims to enhance medication adherence, patient safety, and digital health accessibility for elderly and ill individuals. It serves as a reliable digital

companion that helps users manage their prescriptions, coordinate with family members, and maintain health accountability in a secure and user friendly manner.

The main goals of Medimind are:

- **Precise medication scheduling and alerts:** Users can set specific times for their medications to ensure they never miss a dose, reducing the risk of health complications.
- **Dual mode caregiver synchronization:** The app offers a specialized interface for both patients and monitors, allowing family members to stay informed about the health status of their loved ones via real time synchronization.
- **Offline accessibility and reliability:** The system is designed to provide essential alerts even when the device is offline, ensuring that users are reminded of their medication regardless of internet connectivity.
- **Secure authentication and user login:** Medimind requires a secure login to protect personal schedules, ensuring that only authorized users and their designated caregivers can access the medication plan.
- **Secure data management with Firebase:** Firebase ensures secure authentication and reliable cloud storage, building trust and maintaining the privacy of sensitive health information during the synchronization process.
- **Simple and accessible interface:** The UI is designed specifically for elderly or ill people, featuring a clean layout to reduce confusion and provide a comfortable experience for managing daily pill schedules.

2.3 Purposes

The purpose of developing Medimind goes beyond creating a technical prototype. It aims to bridge the gap between medical requirements and accessible digital support for

elderly and ill individuals. By enabling secure authentication and real time caregiver synchronization, the application seeks to:

- **Encourage health accountability:** Support users in managing their medication schedules and understanding the importance of consistency in their daily routines.
- **Promote medication adherence:** Provide timely alerts and offline notifications to ensure that patients take their prescribed doses at the correct times.
- **Foster a supportive environment:** Create a safe digital space where caregivers and family members can monitor progress and provide necessary assistance to their loved ones.
- **Reduce health risks:** Lower the dangers associated with missed doses or incorrect timing, which is especially critical for chronic illnesses.
- **Enable reliable support:** Act as a preventive tool that ensures constant monitoring, helping to maintain a stable health status through a combination of local and cloud based technology.

Ultimately, Medimind demonstrates how technology can be leveraged reliably to promote safety, independence, and well being for patients and their families.

2.4 Methods

To achieve the objectives of **MediMind**, a systematic and structured development approach was followed:

Requirement Analysis

Observations, discussions, and informal interviews were conducted to understand the needs of elderly and chronically ill users. Special focus was given to medication adherence problems, forgetfulness, usability simplicity, offline accessibility, and caregiver monitoring requirements.

System Design

Technical diagrams including Use Case Diagrams, Entity Relationship (ER) Diagrams, Data Flow Diagrams (DFD), and Activity Diagrams were designed to define the system architecture, data relationships, and workflow processes. The design emphasized simplicity, large readable interfaces, and minimal user interaction complexity to support elderly users.

Implementation

- Flutter and Dart were used to develop a cross-platform mobile application compatible with Android devices.
- Firebase was integrated for secure user authentication, real-time database management, and medicine tracking.
- A notification system was implemented to remind users to take medicines at scheduled times.
- Users can add and delete medicines along with dosage and timing details.
- A monitoring feature was developed allowing up to five caregivers to monitor a patient's medication adherence.
- Offline functionality was implemented so that reminders and stored data remain accessible without continuous internet connectivity.

Testing and Evaluation

Functional testing was conducted for core features such as user registration and login, medicine addition, reminder notifications, monitoring system, and database synchronization. Special usability testing was performed considering elderly users to ensure clarity, readability, and ease of navigation.

Iteration and Refinement

Based on testing outcomes and user feedback, the user interface and overall user experience were refined. Improvements were made to button sizes, color contrast, notification

reliability, and navigation flow to ensure comfort, intuitiveness, and better engagement for elderly and ill users.

2.5 Related Work

Several health applications have inspired the development of Medimind, including Medisafe [1], MyTherapy [2], CareClinic [3] and Mango Health [4]. These applications provide valuable insights into digital health support through features such as medication scheduling, dose tracking, and health logging. However, most of these platforms primarily target general global audiences and focus on comprehensive health management or complex medical reports rather than the specific accessibility needs of the elderly or the collaborative requirements between patients and caregivers.

In contrast, Medimind is designed specifically for elderly and ill individuals along with their monitors. Its unique focus includes:

- **Caregiver visibility:** A dual mode interface allows monitors to link with multiple patients and view a live list of their daily medication status to confirm whether doses were taken or missed.
- **Offline accessibility:** Leveraging the **Flutter** [5] framework, the platform is engineered to trigger medication alerts locally on the device, ensuring reliability even in areas with poor internet connectivity.
- **Localized simplicity:** The user interface is tailored to the specific needs of elderly users, focusing on ease of use and high visibility rather than complex data metrics.
- **Secure Authentication:** Using **Firebase** [6] for the backend, the system ensures that user data is synchronized securely across multiple devices.

While existing applications excel in global scalability and extensive history tracking, Medimind distinguishes itself by focusing on ease of access for the elderly.

Chapter 3: Requirement Specification

This chapter provides a detailed overview of the system requirements that guide the development of the Medimind application. It defines what the system should accomplish and the standards it must maintain to ensure reliability, usability, and performance for patients and their caregivers. The functional requirements describe essential features like medication scheduling, secure login, and offline alerts that fulfill the system goals. Meanwhile, the non functional requirements outline the quality attributes and constraints such as accessibility for elderly users and data security that ensure a smooth and efficient experience.

3.1 Functional

The functional requirements describe the key features and interactions that enable users to engage effectively with the Medimind medication management system. The main functional requirements are as follows:

- **User Authentication:** The system allows users to register, log in, and securely access their personal accounts using email and password verification through Firebase.
- **Medication Scheduling:** Patients can add and manage specific medication names and dosages, setting precise times for reminders to ensure adherence.
- **Offline Alerts:** The application triggers medication reminders locally on the device, ensuring that alerts are delivered even without an active internet connection.
- **Dual Mode Interface:** The system provides distinct interfaces for Patients and Monitors, allowing caregivers to link with specific users to oversee their daily status.

- **Medication Status Viewing:** Monitors can view a real time list of the linked patient status to confirm whether medications for the day have been taken or missed.
- **Admin Control:** Admin users can manage the platform by monitoring total user activity and updating content such as the Frequently Asked Questions and About Us sections.
- **Data Storage and Retrieval:** All medication schedules and user profiles are securely stored in the Firebase database and retrieved efficiently to maintain synchronization between devices.

3.2 Nonfunctional

The non-functional requirements ensure that the Medimind system operates effectively, securely, and reliably under various conditions. These include:

- **Performance:** The system should respond to user actions within two seconds to ensure smooth navigation, especially for time sensitive medication logging.
- **Security:** All user credentials and personal medication schedules must be protected through Firebase encryption to prevent unauthorized access.
- **Usability and Accessibility:** The user interface must be intuitive and specifically designed for elderly or ill people, featuring large icons, clear text, and high contrast for easy navigation.
- **Offline Reliability:** The application must provide a dependable notification system that functions without an active internet connection to ensure user safety at all times.
- **Scalability:** The system must be capable of handling an increasing number of patients and monitors while maintaining synchronization speed across the Firebase backend.

- **Availability:** The system should maintain continuous availability with minimal downtime to ensure that medication data remains accessible to both patients and monitors.
- **Compatibility:** The application should work seamlessly across different Android devices and screen sizes to accommodate the various smartphones used by the target community.

3.2.1 Use Case Diagram

The Use Case Diagram illustrates the primary workflows and interactions between the actors (Patient, Monitor, and Admin) and the Medimind system. Each use case represents a high level task such as user authentication, medicine management, patient tracking, and content administration. Short descriptions of the main use cases are provided below to explain the goals of the actors and how the system supports them. See Figure 3.1 for the Use Case Diagram.

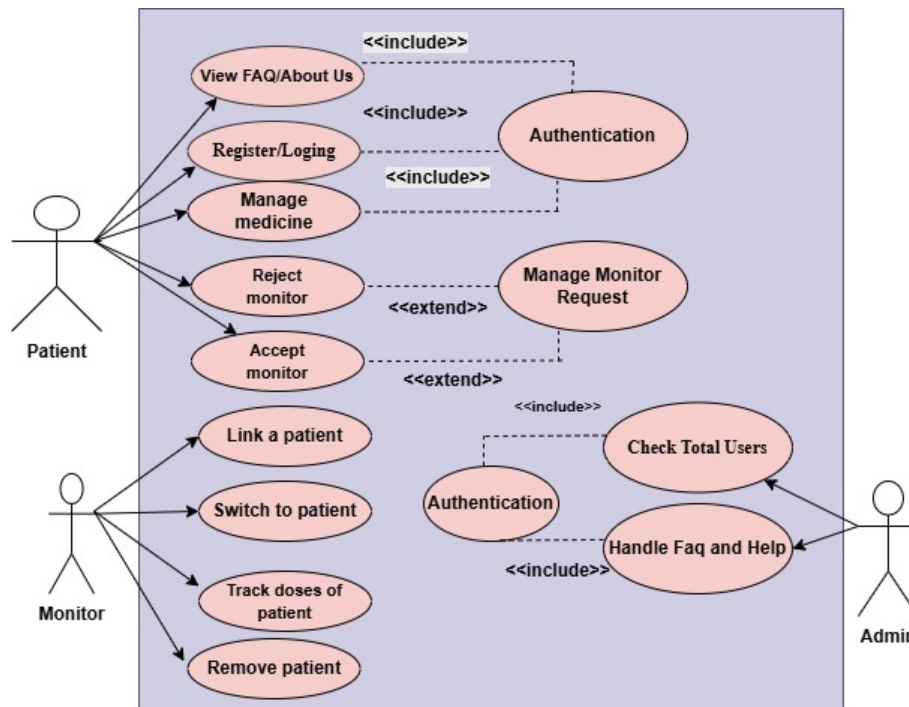


Figure 3.1: Use Case Diagram

3.2.2 Activity Diagram

The Activity Diagram shows the sequence of steps within a selected use case. It maps actions, decision points, and parallel flows to clarify how users and the system progress through tasks. See Figure 3.2 for the Activity Diagram.

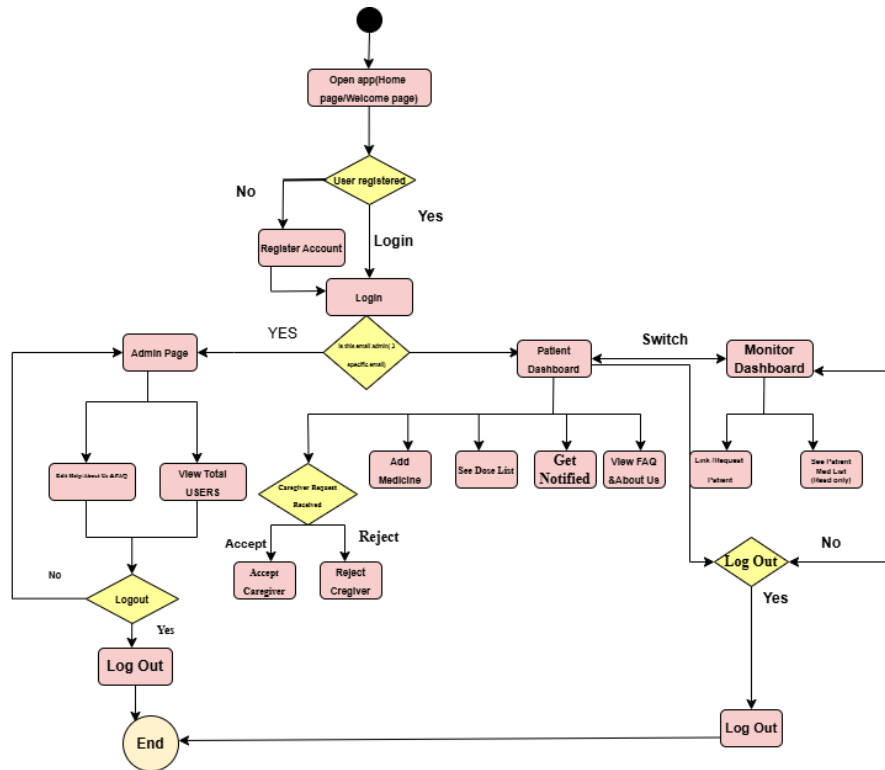


Figure 3.2: Activity Diagram

3.2.3 ER Diagram

The Entity Relationship (ER) Diagram defines the main data entities, their attributes, and relationships. See Figure 3.3 for the ER Diagram.

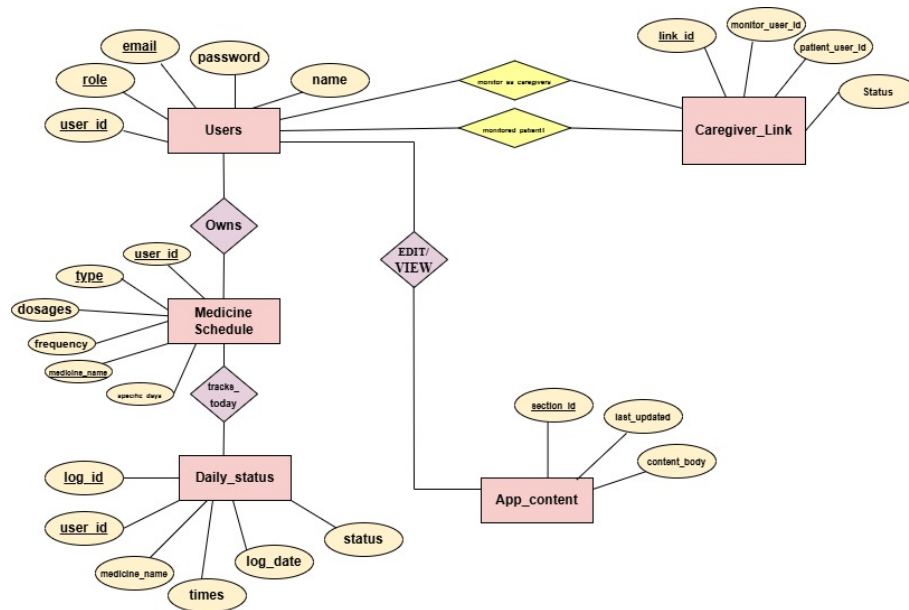


Figure 3.3: ER Diagram

3.2.4 Data Flow Diagram

The Data Flow Diagrams (DFD) illustrate how data moves through the system. DFD Level 0 presents a high-level view of major processes. DFD Level 1 breaks down core processes into sub-processes showing inputs, outputs and data stores. Provide brief descriptions for each process and the data flows linking them. See Figures 3.4 and 3.5.

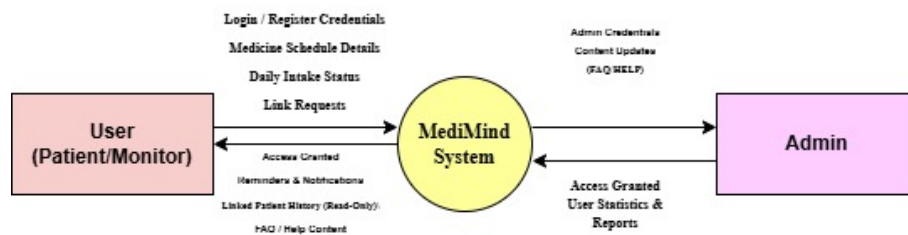
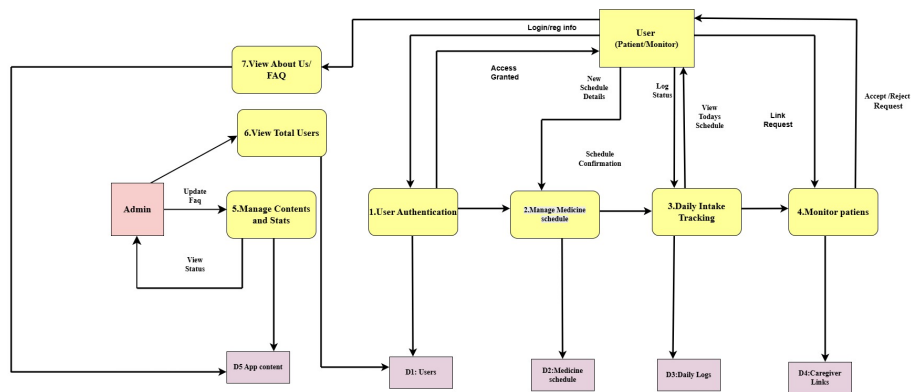


Figure 3.4: DFD Level 0



Chapter 4: Accomplishment

This chapter outlines the steps undertaken to design and develop the Medimind application. It describes how the project was planned, the methods followed during development, and the outcomes achieved at each stage. The goal was to ensure that each component, from requirement gathering to testing, was carried out systematically to meet both the functional and health management objectives of the system.

4.1 Preliminaries

The preliminary phase focused on identifying the problem domain, defining the project scope, and preparing the tools and resources required for implementation. This phase laid the foundation for the development of Medimind as a medication management and caregiver support platform tailored for elderly and ill individuals.

- **Problem Understanding and Research:** The first step involved studying the practical challenges faced by elderly patients and their caregivers in maintaining medication adherence. Research on existing health applications such as Medisafe, MyTherapy, CareClinic, and Mango Health provided insights into current solutions and their limitations. The findings showed that most global apps focused on comprehensive health logging rather than the specific accessibility needs of the elderly or simplified caregiver synchronization.
- **Requirement Gathering:** Informal discussions and interviews were conducted among potential users and their families to understand their health management needs and expectations. The feedback highlighted the importance of simplicity, offline reliability, and secure login. Based on these insights, both functional and non-functional requirements were defined, emphasizing medication scheduling, offline alerts, and a dual mode interface for monitors.

- **Tool Selection:** To ensure an efficient and flexible development process, appropriate tools and technologies were selected:
 - **Frontend:** Flutter and Dart were chosen for cross platform app development due to their rich UI components and smooth performance on Android devices.
 - **Backend:** Firebase was used for secure authentication, real time database management, and cloud data storage.
 - **Offline Logic:** A local notification service was implemented to ensure alerts trigger without an active internet connection.
 - **Design Tools:** Figma was used for creating wireframes and mockups to ensure an intuitive and accessible design for elderly users.
- **Planning and Design:** The development process was divided into several phases like requirement analysis, system design, implementation, testing, and refinement. System models such as Use Case Diagrams, ER Diagrams, Activity Diagrams, and Data Flow Diagrams were prepared to visualize workflows and data structures, ensuring a clear understanding of patient interactions and the monitor portal.

4.2 How the Work Was Done

The development of Medimind followed an iterative and modular approach, where each part of the system was built, tested, and refined before integration.

- **Implementation Phase:**
 - **Frontend Development:** The interface was built using Flutter widgets, ensuring smooth navigation and adaptability across Android devices. Screens such as login, registration, medication scheduling, and the monitor dashboard were implemented with high visibility to ensure accessibility for elderly users.

- **Backend Integration:** Firebase managed authentication, user profiles, and medication schedules. All data transactions like registration, login verification, and real time status updates were securely handled through the cloud database.
- **Offline Logic Integration:** The local notification system was connected with the scheduling engine, processing medication timings and triggering alerts locally on the smartphone even without internet access.
- **Testing and Refinement:** After implementation, functional testing was conducted for features like secure login, medication scheduling, and database synchronization. Specific tests were performed to verify that offline notifications triggered correctly and that the monitor could view the updated status list. Based on initial feedback, UI elements were refined to enhance clarity and ease of use for patients with limited technical backgrounds.
- **Outcome of the Work:** The completed Medimind system successfully achieved its intended goals. It enables users to:
 - Schedule and receive medication reminders accurately via local notifications.
 - Access a secure personal dashboard through a dedicated login system.
 - Provide caregivers with visibility into their daily medication adherence.

The final outcome is a user friendly, reliable, and secure mobile application that promotes safety and independence for elderly and ill individuals.

- **Summary:** In conclusion, the accomplishment phase demonstrates how careful planning, iterative development, and a focus on accessibility contributed to building a functional and supportive digital platform. Medimind stands as an example of how technology can be used reliably to improve healthcare management and support for patients and their families.

Chapter 5: Results

In this chapter, we present the key features and functionalities of our app through screenshots and descriptions. The app provides a user-friendly interface, allowing users to navigate easily between different sections such as welcome page, patient, and monitor dashboard pages.

The screenshots below demonstrate the app's clean layout, where users can access different features efficiently.

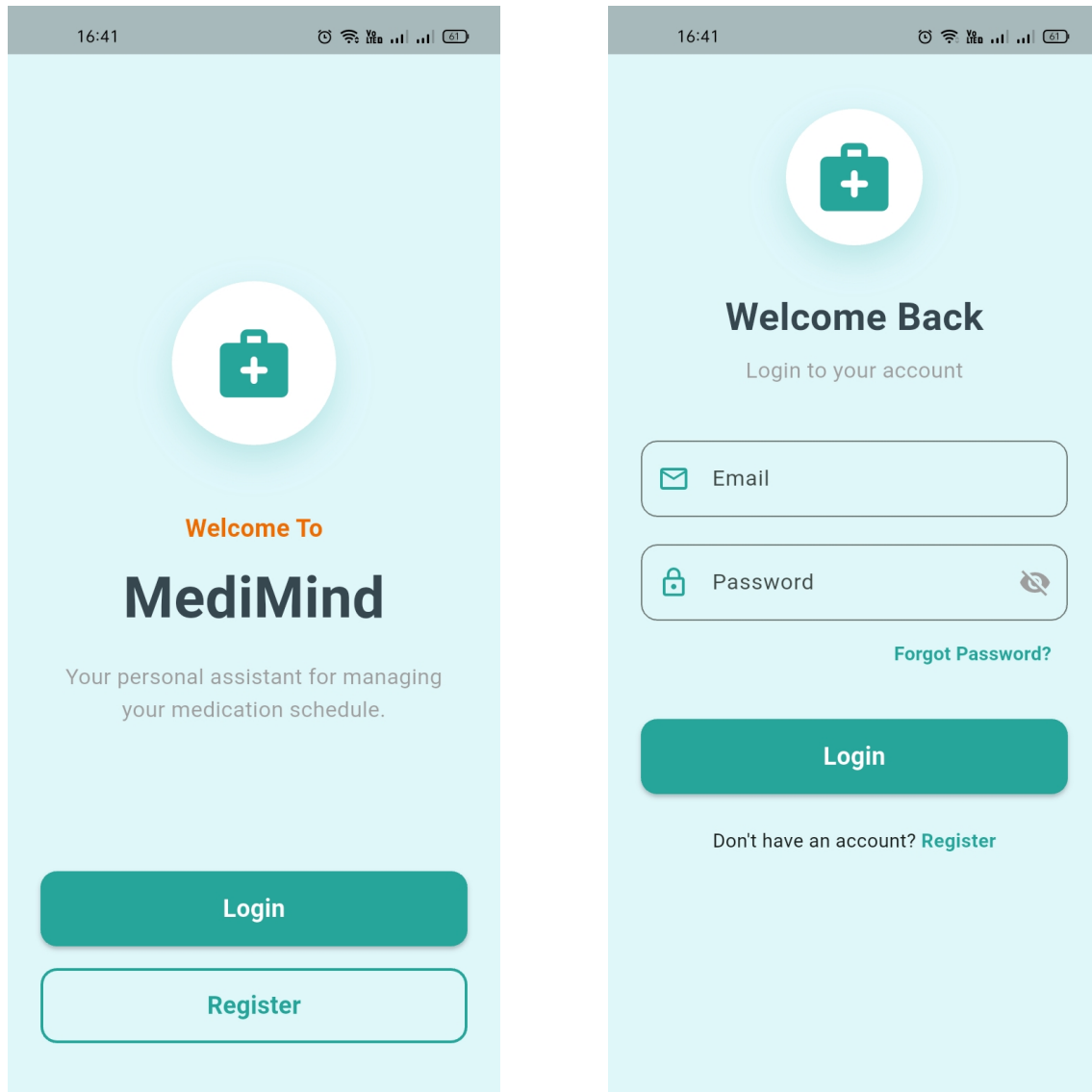


Figure 5.1: Left: Welcome Screen. Right: Login Screen.

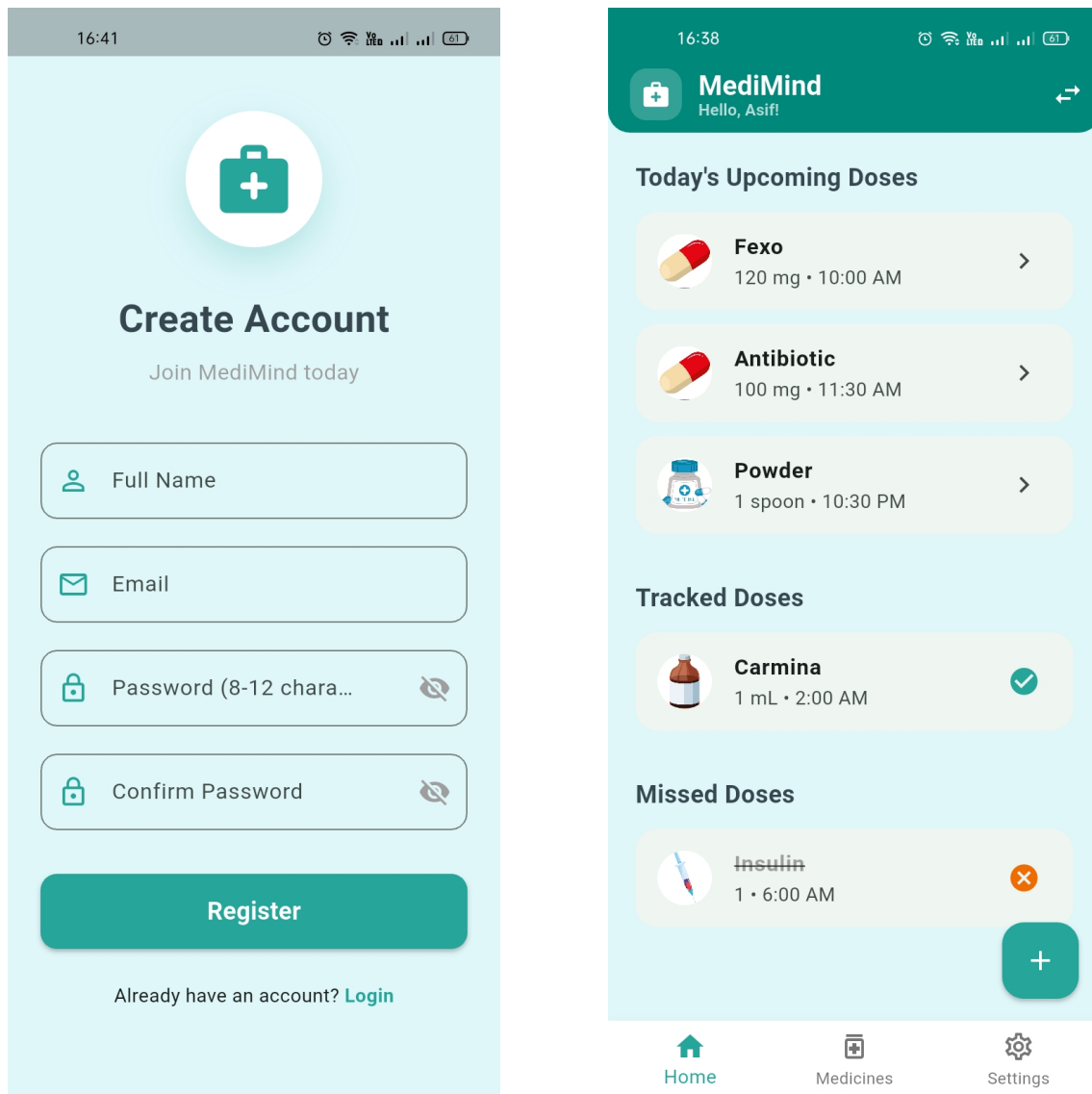


Figure 5.2: Left: Registration Screen. Right: Patient Dashboard Home.

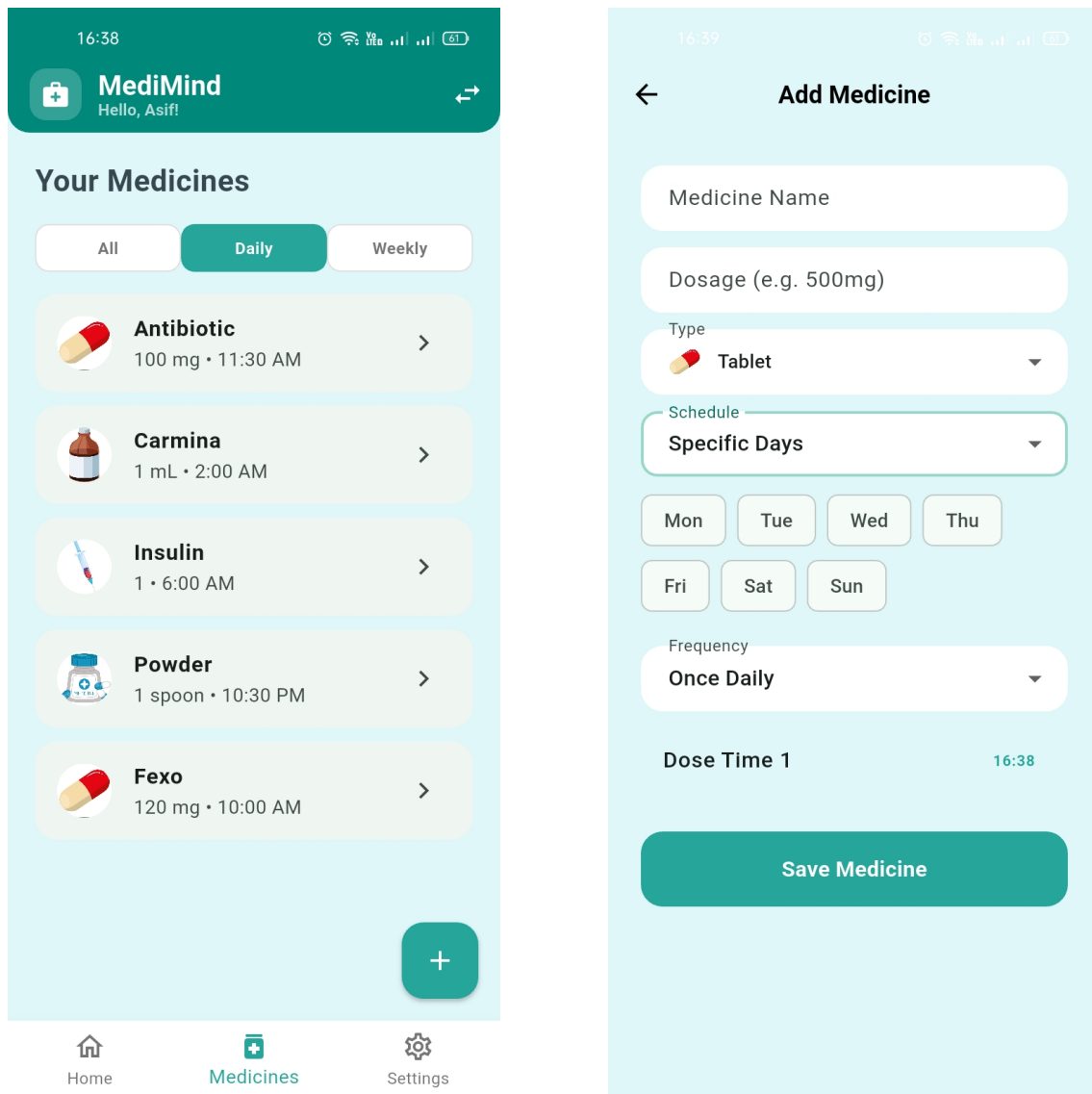


Figure 5.3: Left: My medicine Page. Right: Add Medicine.

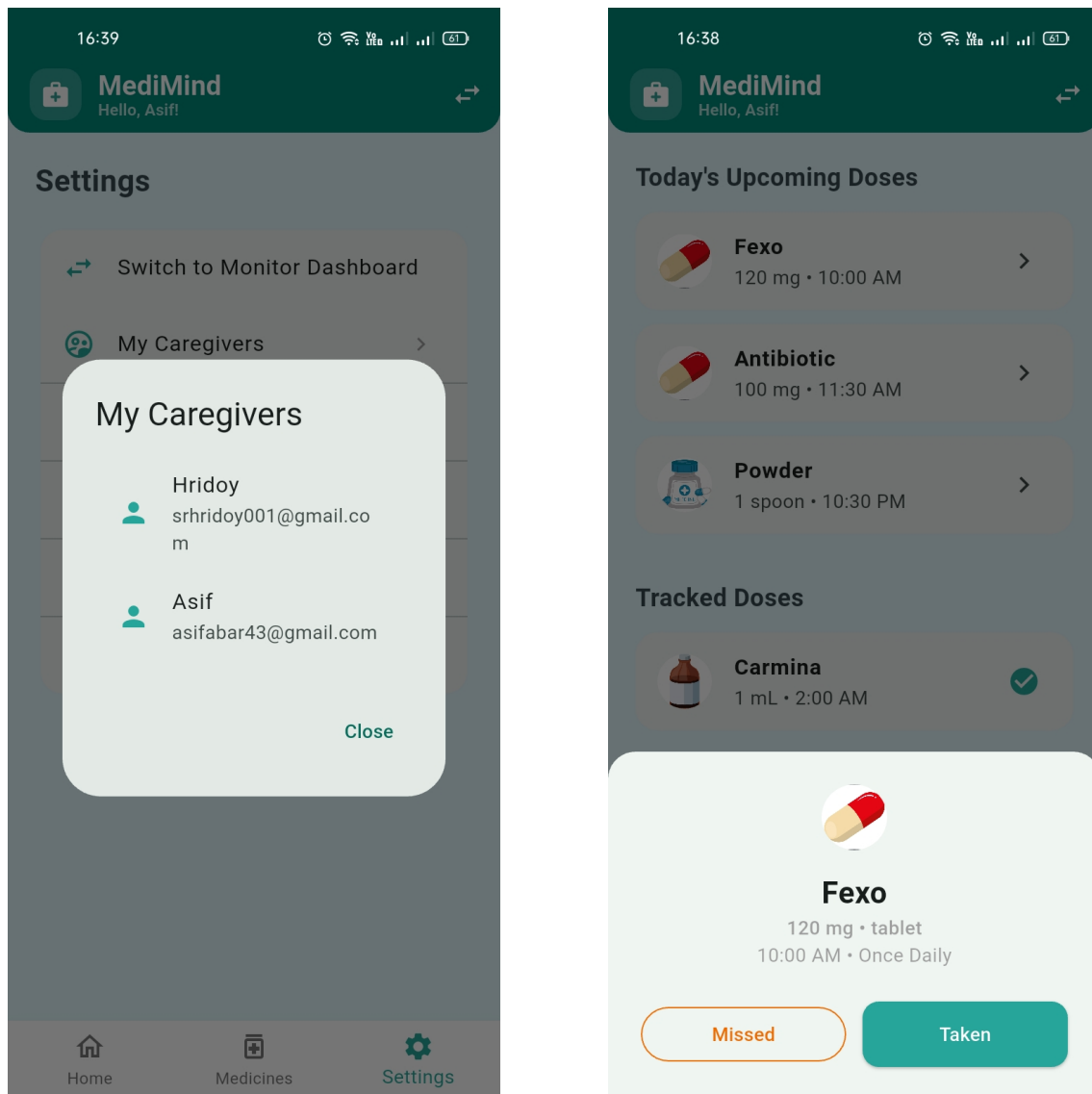


Figure 5.4: Left: Caregiver List. Right: Medicine Take/Missed.

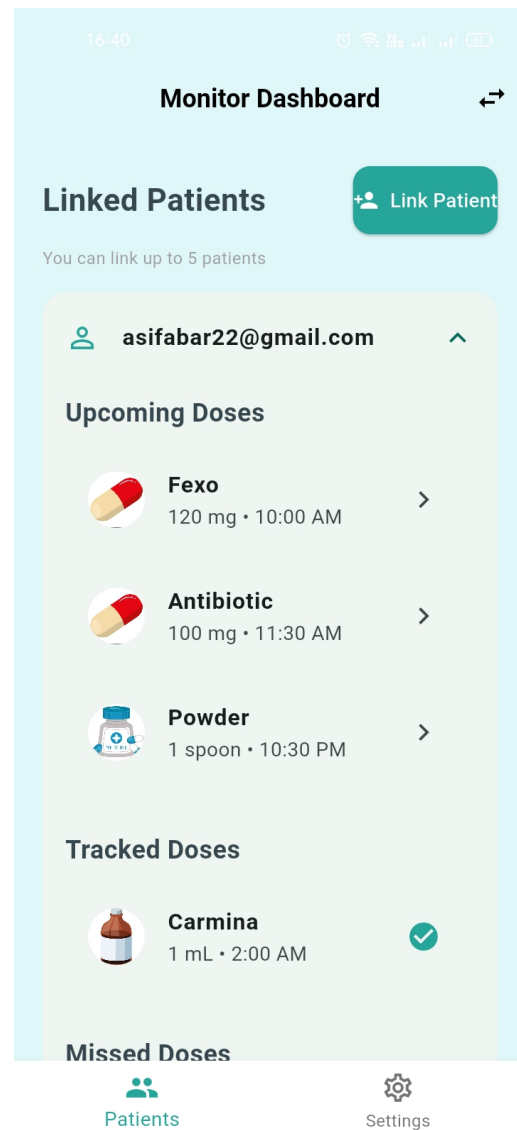
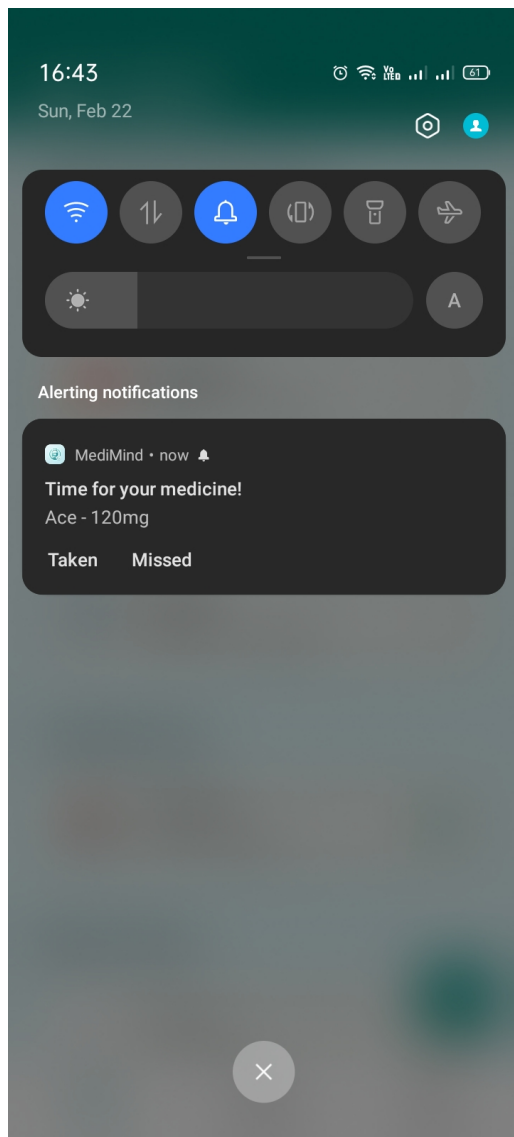


Figure 5.5: Left: Notification. Right: Monitor Dashboard Home.

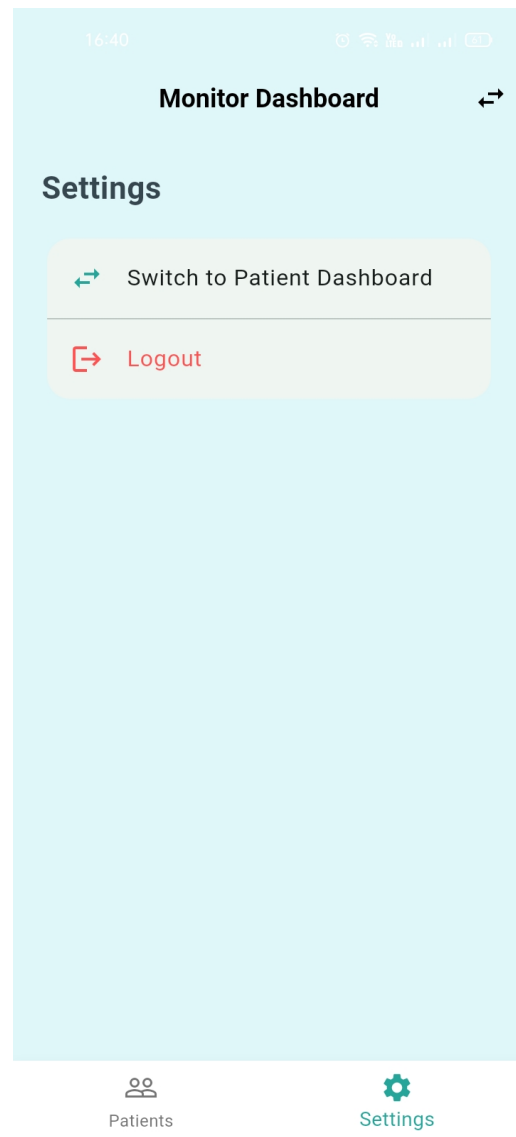
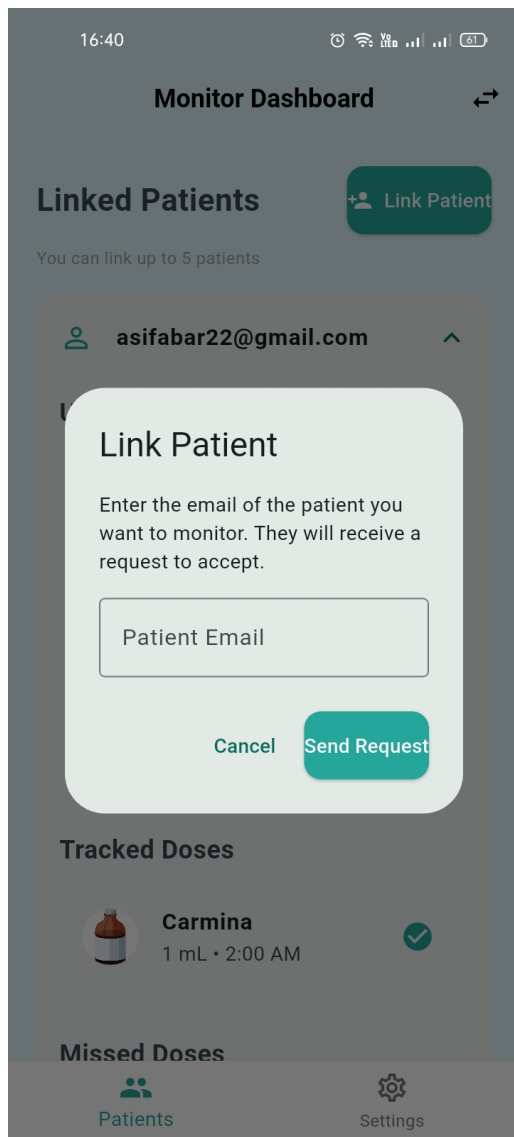


Figure 5.6: Left: Link A Patient. Right: Monitor Dashboard Setting.

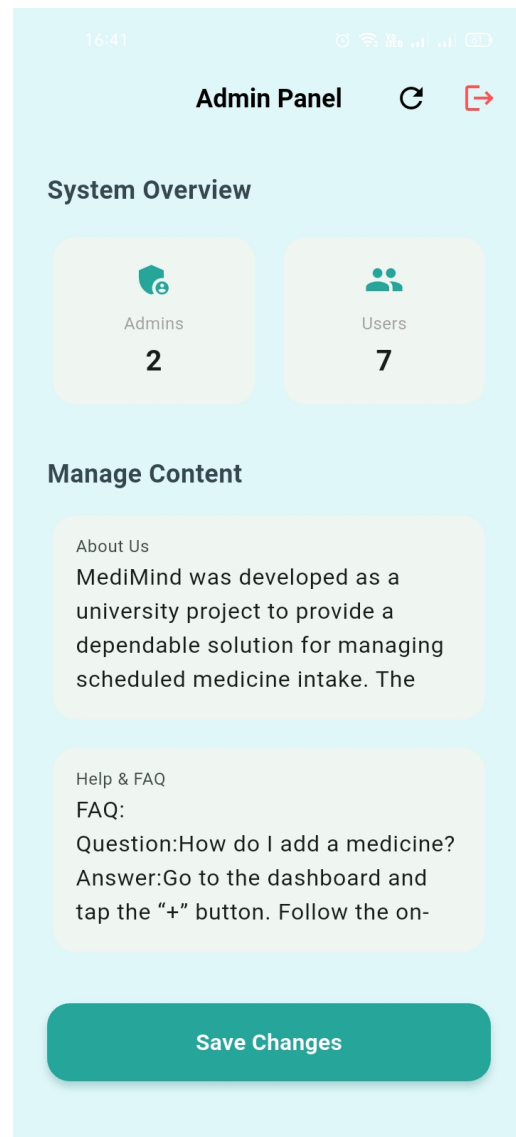
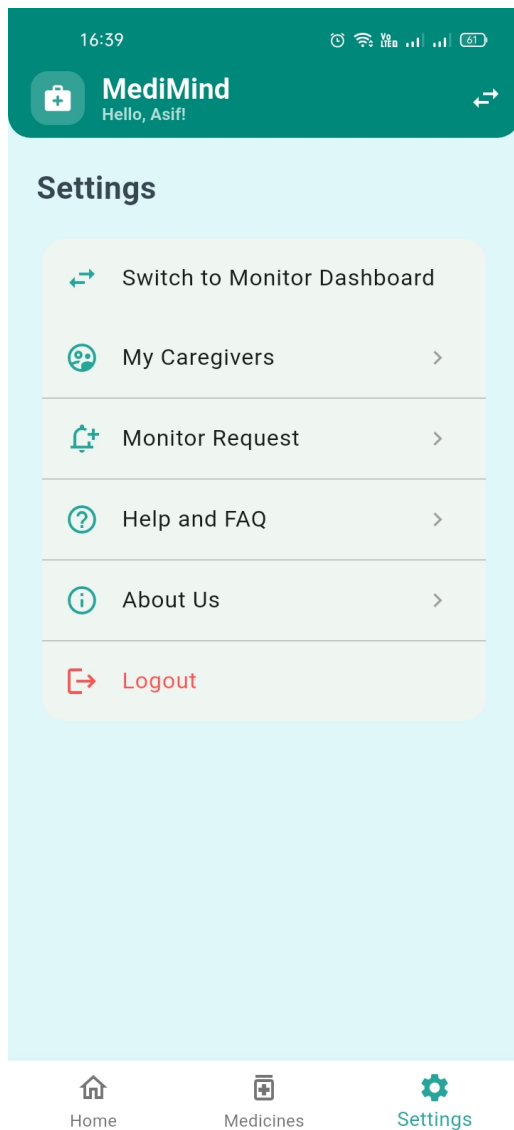


Figure 5.7: Left: Patient Dashboard Setting. Right: Admin Panel.

Chapter 6: Conclusion

The development of Medimind represents an important step toward creating a localized, user centered digital platform that promotes medication adherence and health accountability among elderly and ill individuals. The project successfully met its primary objectives by integrating precise medication scheduling, a dual mode interface for caregivers, secure Firebase authentication, and reliable offline notification capabilities into one cohesive and user friendly mobile application.

While the process involved multiple challenges, each stage from requirement analysis to design, implementation, and testing offered valuable insights into the technical dimensions of building a health oriented system for users with specific accessibility needs. The result is a working prototype that demonstrates how technology can support safety and independence for patients and their families in a secure digital environment.

6.1 Restrictions

During the development process, several constraints shaped the project scope and outcomes:

- **Restricted Database Access:** The project was limited by the lack of premium access to database services, which influenced data storage capacity and synchronization speeds.
- **Limited Alarm Functionality:** Due to technical and resource constraints, a specialized high priority alarm mode could not be integrated into the current version of the application.
- **Connectivity Dependency:** While essential reminders function offline, the system requires active internet connectivity for core backend operations and data synchronization via Firebase.

- **Device Compatibility:** Testing and optimization were restricted to Android platforms, limiting the immediate availability of the system for iOS users.
- **Time and Resource Constraints:** The development timeline prevented more extensive user interface enhancements and the inclusion of advanced features like long term medical history.

These restrictions influenced the overall performance and generalizability of the system but also clarified key areas for future development and optimization.

6.2 Limitations

Despite achieving its functional goals, the Medimind system still has several limitations that define the boundaries of its current version:

- **Restricted Database Resources:** The project was developed without premium database access, which limited the capacity for large scale data management and complex storage operations.
- **Alarm Mode Constraints:** Due to technical limitations during the implementation phase, a dedicated high priority alarm mode could not be added to the notification system.
- **Lack of Medical History:** The current version focuses on daily medication status and does not include a comprehensive long term medical report or history feature.
- **Platform Limitation:** The application is currently optimized and tested for Android smartphones only, lacking compatibility for other mobile operating systems.
- **Simplified Synchronization:** While the system supports caregiver viewing, it does not currently provide push notifications to the monitor, who must manually check the status list.

These limitations do not diminish the system overall value but rather highlight opportunities for technological and design improvements in future iterations.

6.3 Future Work

Looking ahead, the Medimind project can evolve into a more intelligent, inclusive, and impactful platform through several planned enhancements:

- **Integrate Advanced Alarm Modes:** Future iterations could include high priority alarm modes that bypass silent settings to ensure medication is never missed.
- **Develop Medical History and Reporting:** Building a comprehensive history feature would allow patients and caregivers to track long term health trends and generate medical reports.
- **Expand System Compatibility:** The goal is to extend accessibility by developing versions for iOS and web platforms.
- **Implement Real Time Monitor Notifications:** Future updates could include push notifications for caregivers when a patient misses a dose, rather than relying on manual list checks.
- **Secure Premium Database Access:** Moving to a premium backend tier would allow for larger data storage, faster synchronization, and more complex user management.
- **Continuously Improve UI and Accessibility:** Based on feedback from elderly users, the interface will be further refined to ensure the highest level of ease and comfort.

In building Medimind, we set out to turn digital tools into health fields like helping elderly and ill people manage, track, and act on their medical needs. While our journey uncovered both meaningful insights and inevitable setbacks such as limited resources and

technical constraints, the core promise remains unchanged: medication adherence is the quiet foundation of safer health, independent living, and stronger well being.

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Appendix A: Source Code

This appendix provides a brief overview of supplementary materials related to the Medimind system. It summarizes additional notes, references, and technical resources that support the project but are not included in the main chapters.

Future versions of this report may include extended sections such as:

- **Source code excerpts:** Implementation examples focusing on the Flutter frontend and the Firebase backend integration.
- **User manuals:** Comprehensive setup guides designed for patients, monitors, and admin users to ensure easy navigation for elderly individuals.
- **Technical documentation:** Detailed explanations of the offline notification logic and the database synchronization architecture.